

AI & IoT-powered smart home energy management application

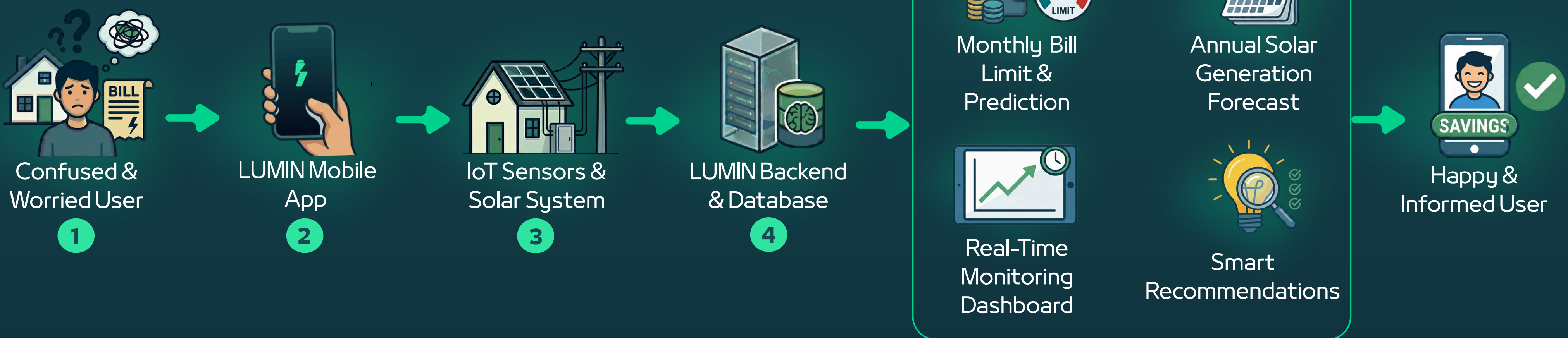
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PROBLEM

Electricity consumption in Saudi Arabia ranks among the world's highest per household according to Matar and Anwer's 2017 Energy Policy study, yet homeowners lack the tools to identify which devices drive their bills, reduce their consumption, or evaluate solar investment. The result is wasted energy and missed savings

OBJECTIVE

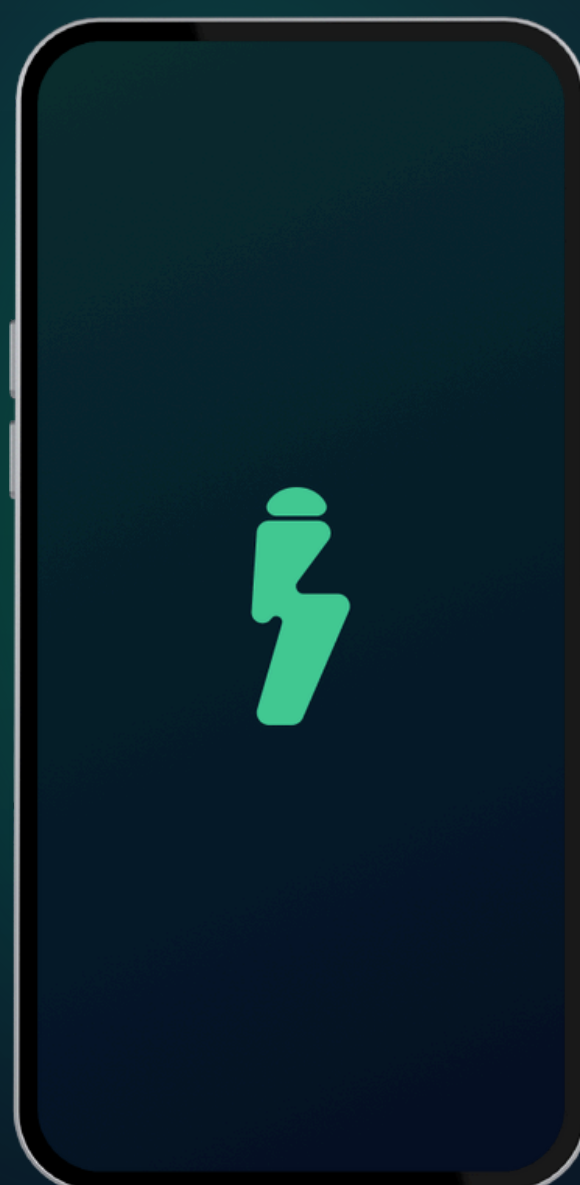
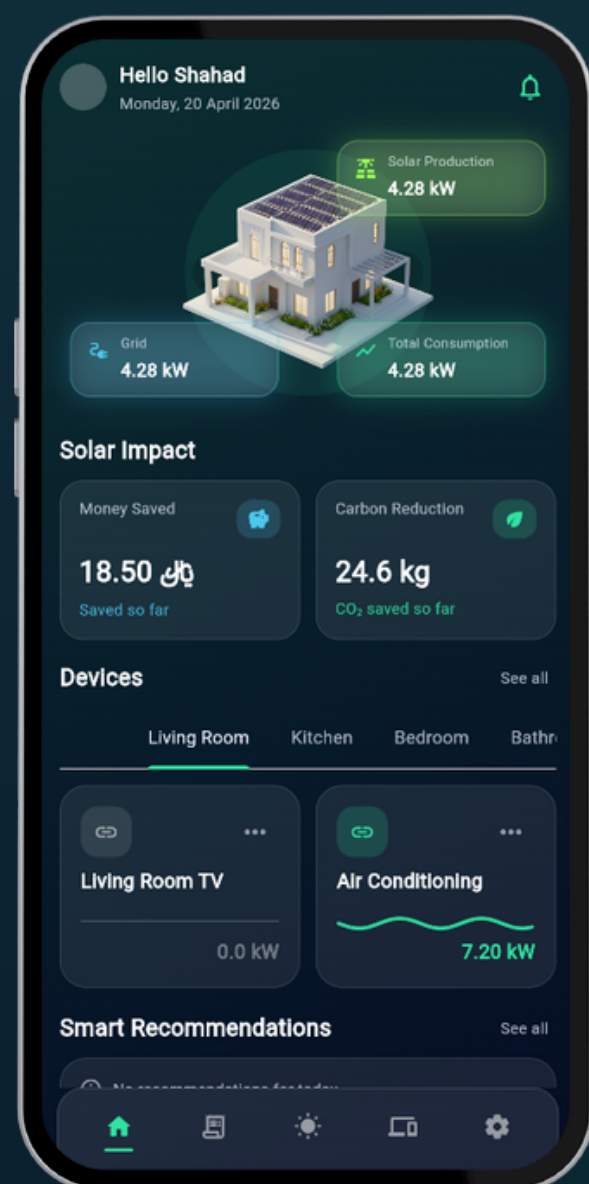


SOLUTION

The proposed system is an IoT-powered smart home energy management application that connects to your devices and solar panels, delivering real-time energy monitoring, bill limit control, and AI-driven solar forecasts to reduce consumption and optimize investment decisions.

Real-Time monitoring

- Monitor real-time energy consumption per device
- Identify high-consuming devices and connection issues instantly
- Track your total energy savings

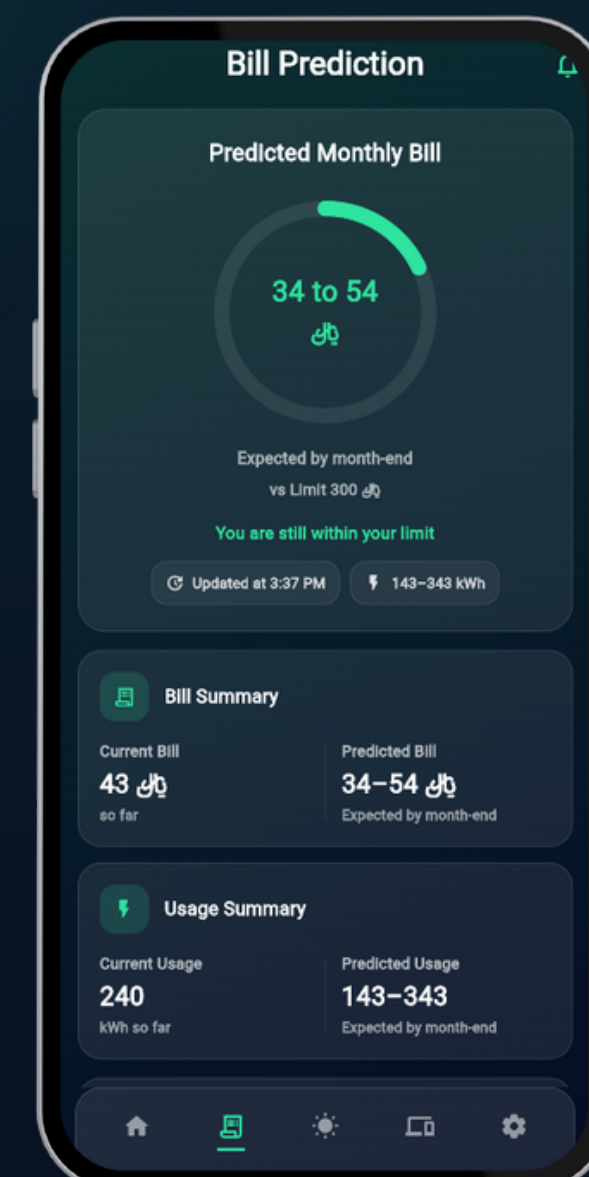
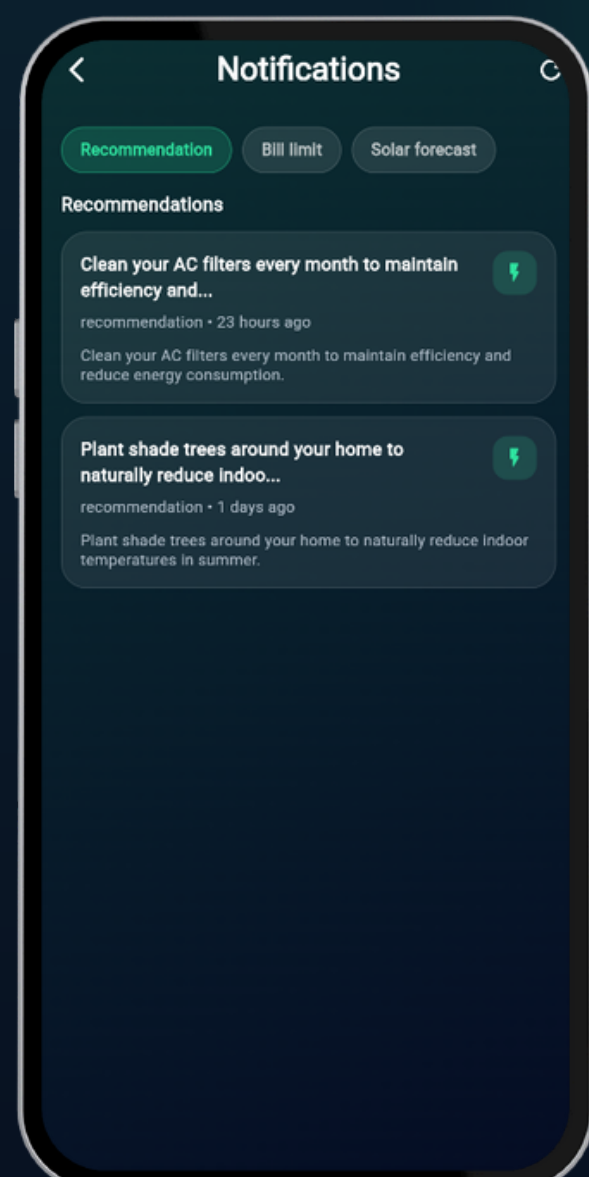


Solar Forecast

- Forecast solar production over the next two years with 97% XGBoost accuracy
- Track expected energy production in kWh for your household
- Get an investment recommendation based on your solar returns

Smart Recommendations

- Get energy-saving tips based on your consumption
- Discover the best times to run appliances using solar insights
- Stay informed with smart alerts



Bill Limit & Prediction

- Set a monthly bill limit with automatic weekly alerts
- Receive weekly predictions of your end-of-month bill
- Stay within budget and avoid unexpected bills

TOOLS

